Section B1: Analysis of mymart_debug.py

Although the model reports an impressive test accuracy of 97%, this result is misleading when viewed in the context of fraud detection. Since chargeback transactions make up only a small percentage of all transactions, a model can achieve very high accuracy while still failing to identify most fraudulent cases. After reviewing the code, three major issues were identified that explain why the model performs poorly in detecting chargebacks.

Issue 1: Data Leakage During Scaling

One significant problem is that the numerical features are scaled before the dataset is split into training and testing sets.

df[numeric_cols] = scaler.fit_transform(df[numeric_cols])

This means the scaler learns information from the entire dataset, including the test data that should remain unseen until evaluation. As a result, the model indirectly gains knowledge about the test set during training, creating a situation known as data leakage.

For MyMart, this leads to an overly optimistic view of model performance because the evaluation is no longer based on completely unseen data. Consequently, the reported test accuracy and other evaluation results may appear better than they would be in a real-world deployment.

Issue 2: The Class Imbalance Problem Is Ignored

The code also does not address the fact that chargebacks are relatively rare compared to legitimate transactions. Fraud detection datasets are typically highly imbalanced, meaning most records belong to the non-fraud class.

Without techniques such as class weighting, oversampling, or undersampling, the model naturally learns to favor the majority class. In practice, this means it may predict almost every transaction as legitimate and still achieve a high accuracy score.

This is particularly harmful for MyMart because the business goal is not simply to classify transactions correctly overall, but to identify fraudulent transactions before losses occur. The issue mainly affects the confusion matrix, recall score, and the number of fraud cases successfully detected. A high accuracy score may therefore hide poor fraud detection performance.

Issue 3: Accuracy Is the Only Evaluation Metric Used

The model is trained and evaluated using accuracy as its only performance metric.

metrics=['accuracy']

While accuracy is easy to understand, it is not suitable for highly imbalanced problems such as chargeback detection. A model can achieve excellent accuracy while missing most fraudulent transactions, which is exactly the concern raised by the operations manager.

More informative metrics such as recall, precision, F1-score, ROC-AUC, and the confusion matrix should be used to evaluate how effectively the model identifies fraud. These measures provide a clearer picture of business risk and model effectiveness.

Conclusion

The model's reported 97% accuracy creates a false sense of confidence. Data leakage during scaling, failure to handle class imbalance, and an overreliance on accuracy as the sole evaluation metric all contribute to misleading performance results. As a consequence, the model may appear successful while missing a large proportion of chargeback transactions. Addressing these issues is essential if MyMart wants a fraud detection system that genuinely protects revenue and reduces financial losses.